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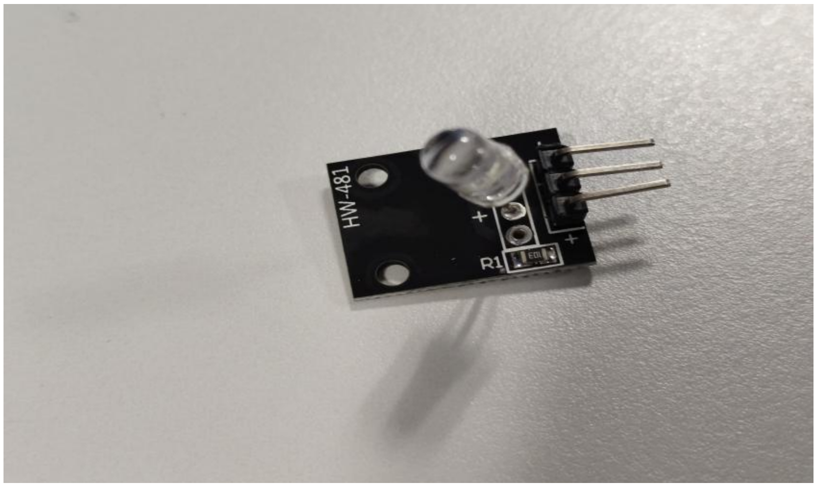
Day 7 - In-Depth Study of Arduino-Compatible HW Series Sensor Modules

1. 7-Color Flashing LED Module (HW-481)



What is the Sensor?
The HW-481 is a self-contained LED module that cycles through 7 colors automatically using an in-built IC. It is not a "sensor" per se, but rather an output device used for visual effects.

Why is it Used?

- Ideal for decoration, light shows, or indicating status.
- Used in DIY projects to visually alert or entertain users.

Pin Details:

- VCC (Power, typically 5V)
- GND (Ground)

How to Connect:
Connect the VCC pin to 5V on the Arduino and the GND pin to Arduino GND. No data input is needed.

Advantages:

- Easy to use, no code needed.
- Attractive for beginner projects.

Disadvantages:

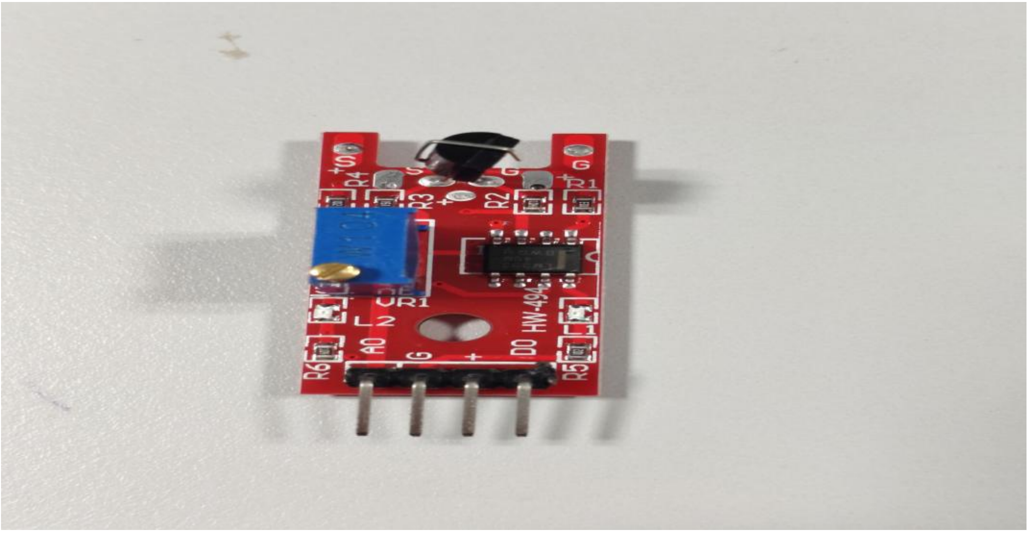
- Cannot control colors manually.

- Limited use beyond decoration.

Limitations:

- Color transitions are predefined.
- No programmability or interactivity.

2. Flame Detection Sensor Module – HW-494



What is the Sensor?
HW-494 is an IR-based flame detection module. It can detect flames or light sources with a wavelength of 760 nm to 1100 nm.

Why is it Used?

- Fire detection systems
- Safety alarms in industrial or home environments

Pin Details:

- VCC
- GND
- DO (Digital Output)
- AO (Analog Output)

How to Connect:

- VCC to 5V, GND to GND

- DO to a digital input on Arduino
- AO to an analog input for intensity measurement

Advantages:

- Detects flame up to 80cm depending on brightness
- Digital and analog output

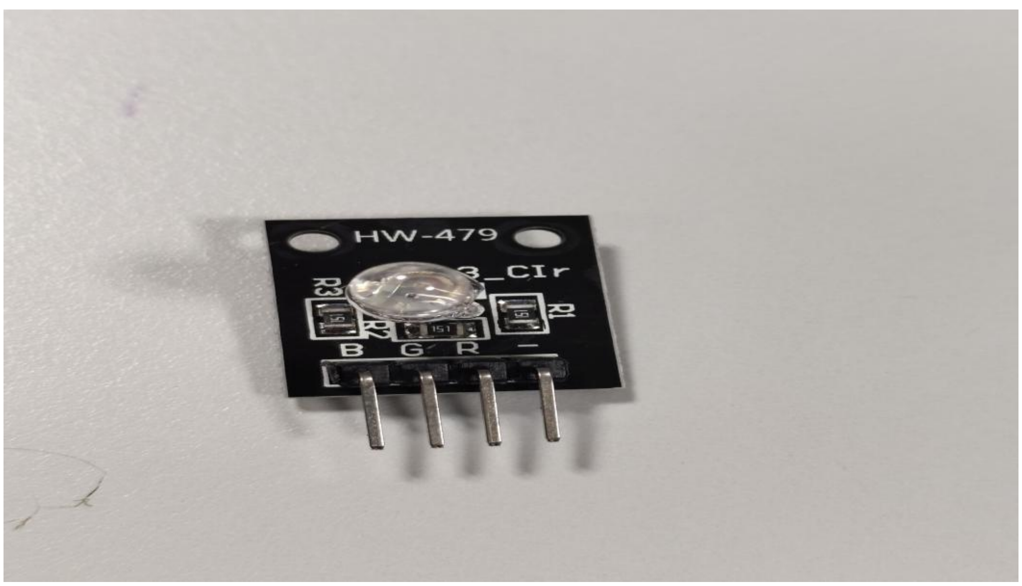
Disadvantages:

- False positives in presence of sunlight or IR sources

Limitations:

- Needs proper calibration
- Can’t differentiate flame sources

3. RGB LED Module – HW-479



What is the Module?
An RGB LED module that allows the user to create various colors by adjusting the brightness of Red, Green, and Blue LEDs.

Why is it Used?

- Status indicators
- Custom lighting effects

- Interactive visual feedback

Pin Details:

- R (Red)
- G (Green)
- B (Blue)
- Common Cathode/GND

How to Connect:

- Connect R, G, B pins to PWM-enabled Arduino pins
- Connect common GND to Arduino GND

Advantages:

- Full control over color via PWM

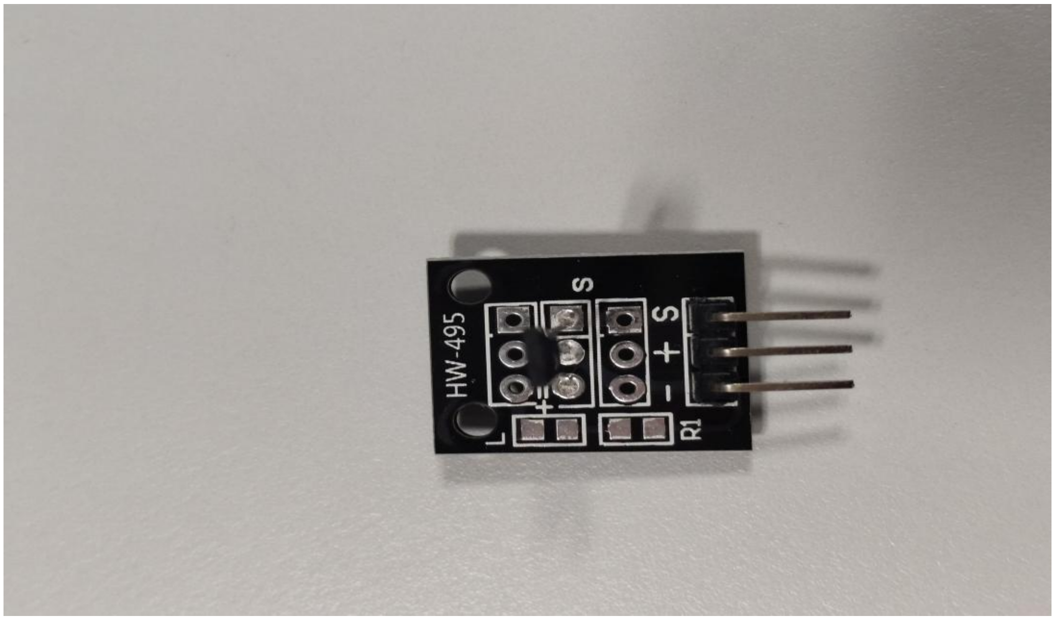
Disadvantages:

- Requires multiple pins
- Brightness varies with current

Limitations:

- Can consume more power
- Limited viewing angle

4. Analog Hall Effect Sensor Module – HW-495



What is the Sensor?
HW-495 uses the Hall Effect to detect magnetic fields. It produces an analog voltage that varies with magnetic field strength.

Why is it Used?

- Detecting magnetic poles
- Proximity sensing
- Tachometers

Pin Details:

- VCC
- GND
- AO (Analog Output)

How to Connect:

- VCC to 5V
- GND to GND
- AO to analog pin (A0)

Advantages:

- Contactless sensing
- Durable and long-lasting

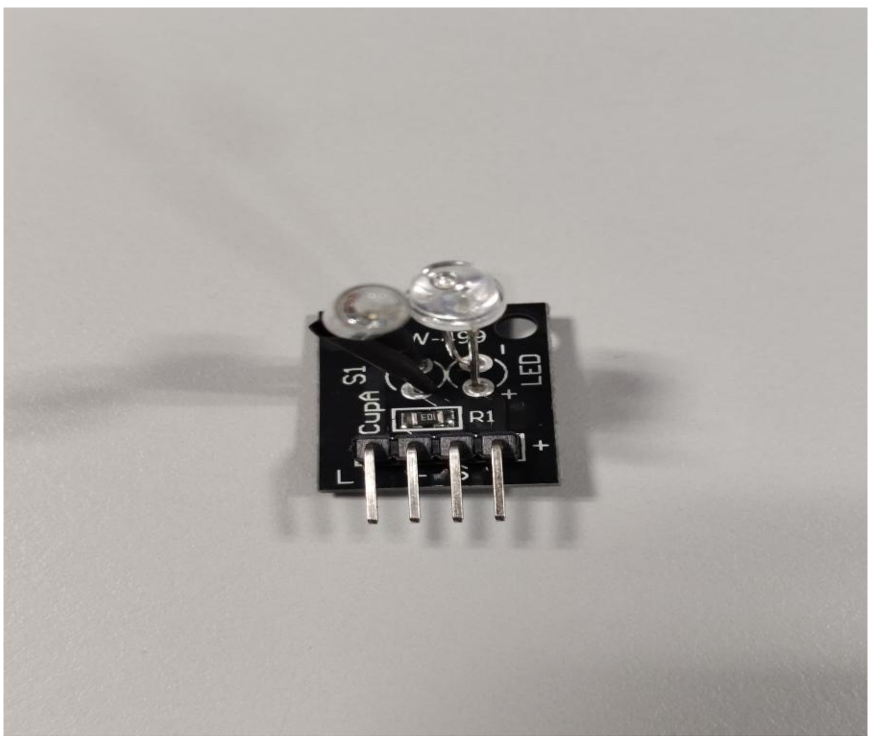
Disadvantages:

- Sensitive to strong magnetic noise

Limitations:

- Non-linear response
- Limited range

5. Magic Light Cup Module – HW-499



What is the Module?
It simulates a "magic" light cup that turns on/off based on orientation, typically using a mercury tilt switch and LED.

Why is it Used?

- Fun DIY educational kits
- Interactive visual feedback

Pin Details:

- VCC
- GND

How to Connect:

- Connect to 5V and GND of Arduino
- Use external transistor for control if needed

Advantages:

- No programming needed
- Responds to movement

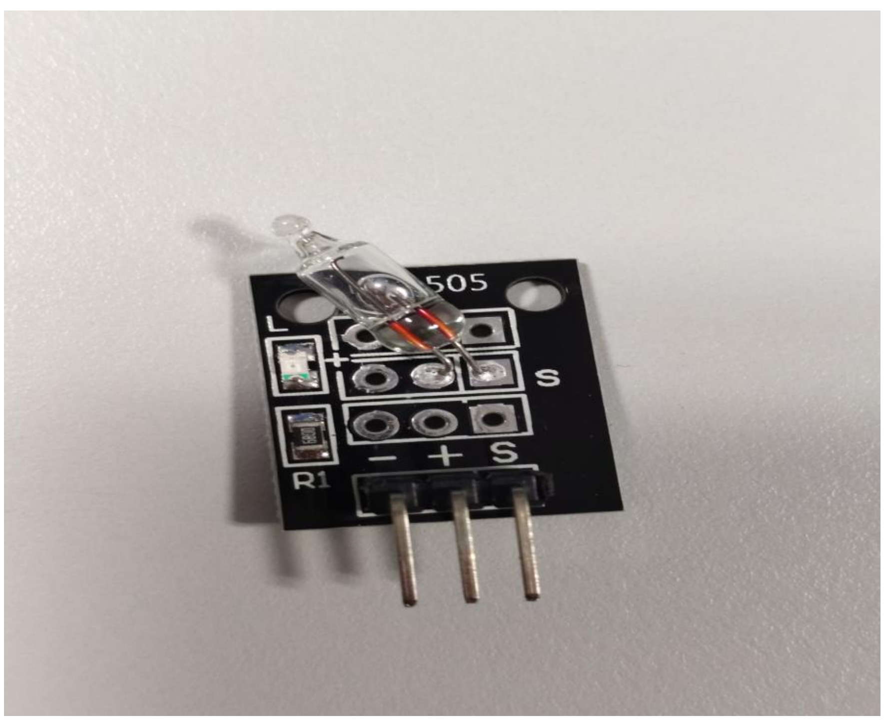
Disadvantages:

- No precise control or input

Limitations:

- Simple novelty device, not for professional use

6. Mini PIR Motion Sensor Module – HW-505



What is the Sensor?
A mini PIR sensor detects motion using infrared radiation emitted by humans or animals.

Why is it Used?

- Home automation
- Intruder detection

Pin Details:

- VCC (3.3–5V)
- GND
- OUT (Digital Output)

How to Connect:

- VCC to 5V, GND to GND
- OUT to a digital input pin on Arduino

Advantages:

- Low power
- High sensitivity to movement

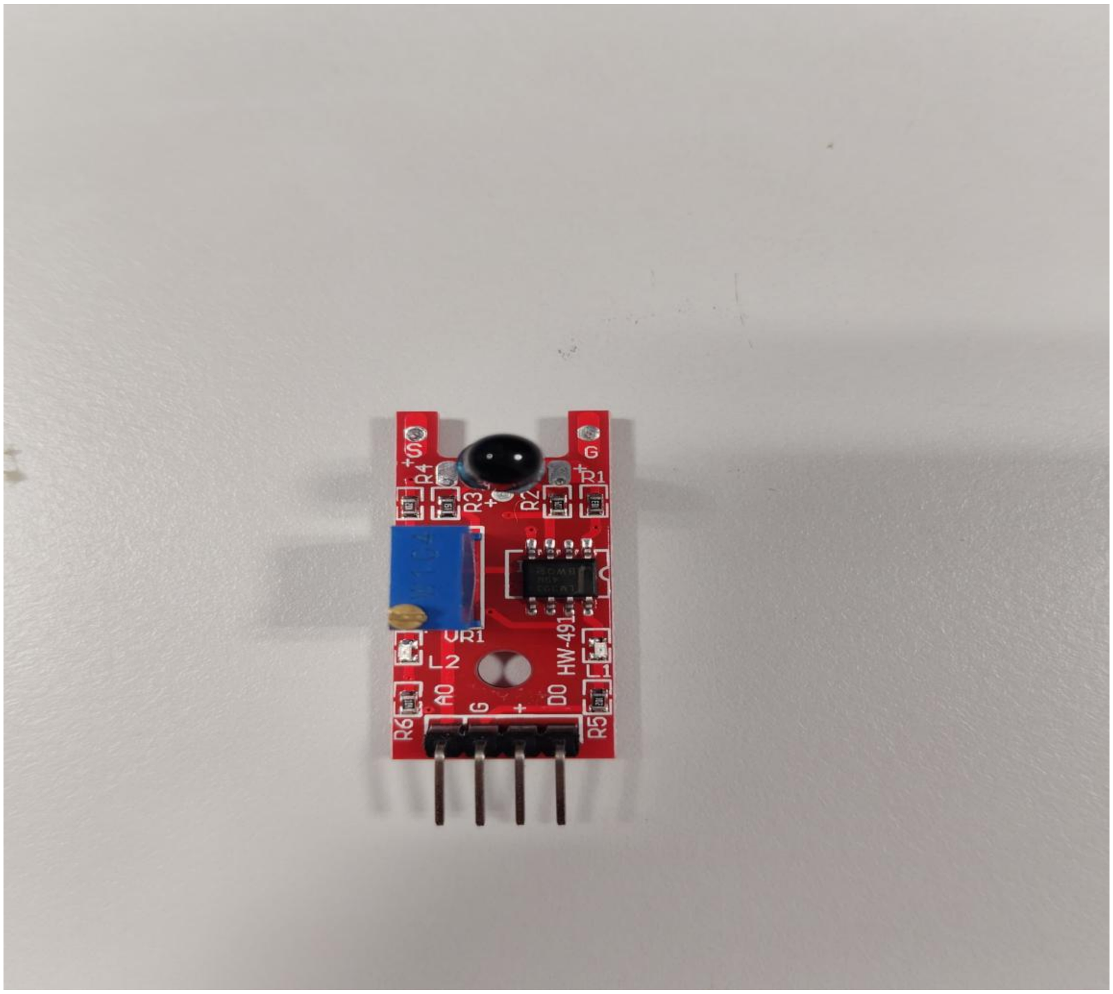
Disadvantages:

- False triggers from temperature changes

Limitations:

- Needs warm-up time
- Detection angle is limited

7. IR Obstacle Avoidance Sensor – HW-491



What is the Sensor?

Uses an IR transmitter and receiver to detect obstacles based on reflected IR light.

Why is it Used?

- Robot navigation
- Object detection

Pin Details:

- VCC
- GND
- OUT (Digital Output)

How to Connect:

- VCC to 5V, GND to GND

- OUT to a digital pin on Arduino

Advantages:

- Affordable
- Good for short-range sensing

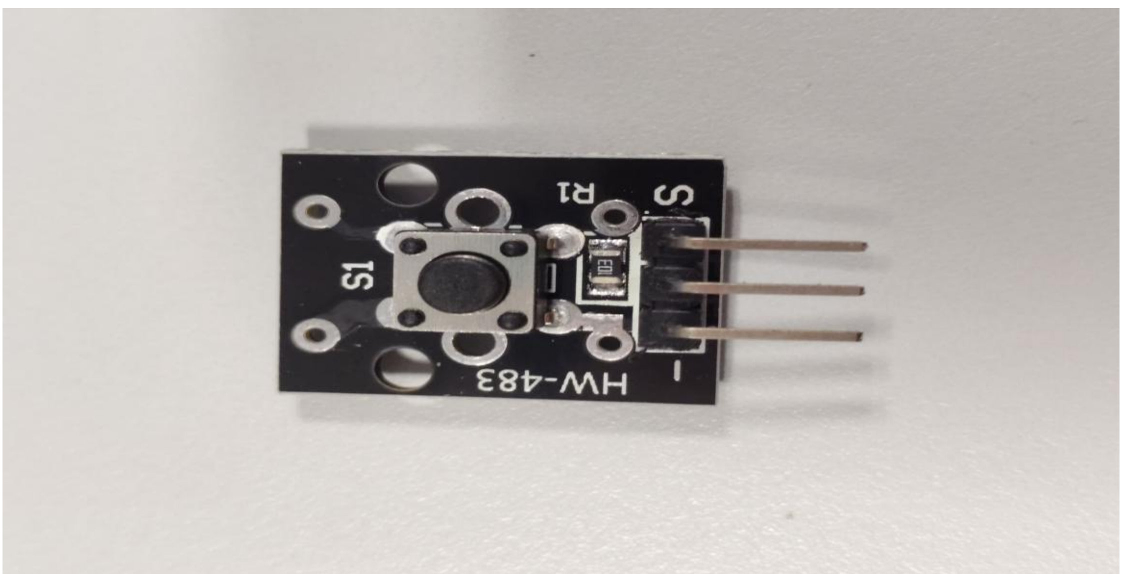
Disadvantages:

- Affected by ambient light

Limitations:

- Short detection range (2–30 cm)

8. Push Button Module – HW-483



What is the Module?
Simple tactile push button used to provide user input to Arduino

Why is it Used?

- User interface
- Manual triggers in circuits

Pin Details:

- SIG (Signal)
- VCC

- GND

How to Connect:

- Connect SIG to a digital pin
- VCC to 5V and GND to GND

Advantages:

- Easy to use
- Provides real-time control

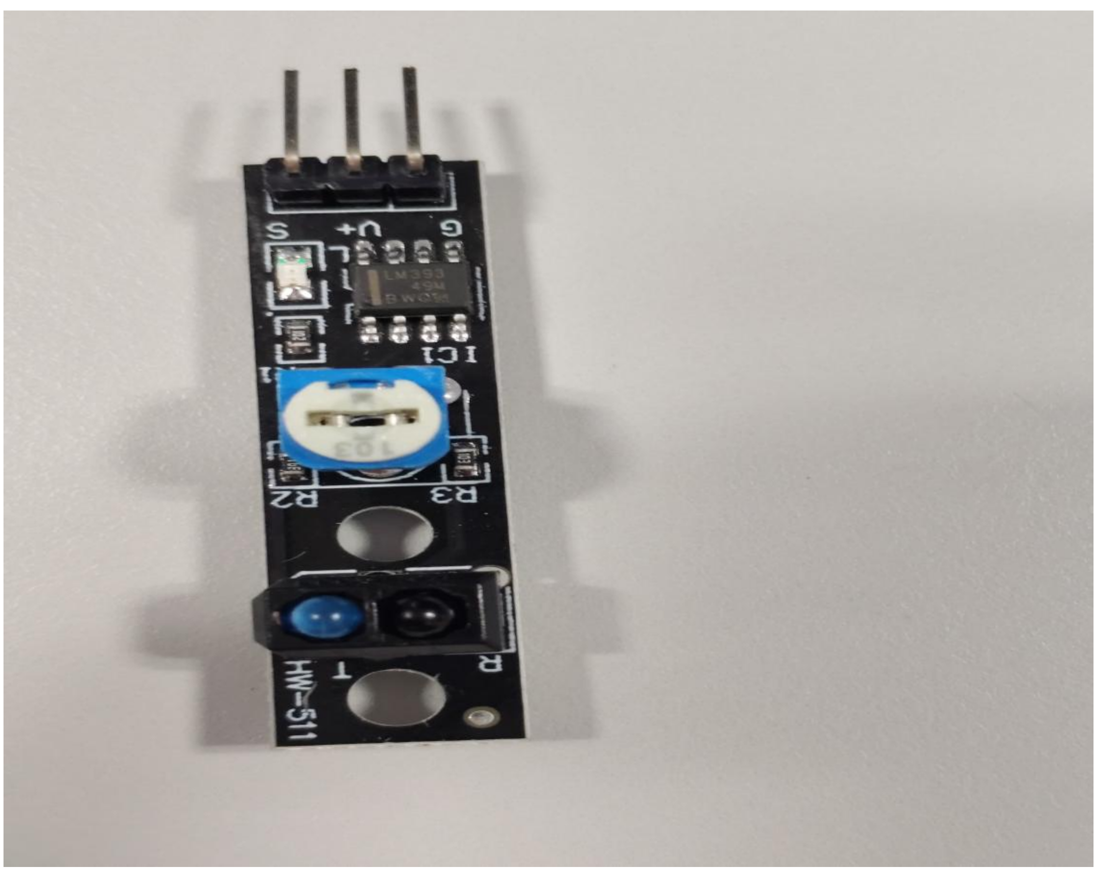
Disadvantages:

- Debouncing required in code

Limitations:

- Mechanical wear over time

9. TCRT5000 IR Line Tracking Sensor – HW-511



What is the Sensor?
TCRT5000 uses infrared to detect black and white surfaces, often used in line-following robots.

Why is it Used?

- Line tracking
- Edge detection

Pin Details:

- VCC
- GND
- DO (Digital Output)
- AO (Analog Output)

How to Connect:

- VCC to 5V, GND to GND
- DO to digital input, AO to analog input

Advantages:

- Accurate for line following

Disadvantages:

- Can be affected by surface reflectivity

Limitations:

- Works only under proper lighting and surface contrast